1. You are the Product lead for TikTok Feed Quality and when you come in to work on a certain day all your teams' key metrics are tanked. How will you go about finding the root cause?

Step 1: Clarify the Problem Scope

Before jumping into solutions,

- **When** did the drop start? Sudden (past 24 hours) or gradual?
- Is the drop **global or localized** (region, platform, version)?
- Which user segments are most affected (Gen Z, Android users, creators)?
- Was there any recent **deployment**, **ranking model change**, or **infra update**?
- Are we confident it's **not a tracking/ETL failure**?
- What are the **specific metrics** that dropped? (e.g., Avg Watch Time, Like Rate, Retention, FYP Engagement Score, Satisfaction Surveys)

Step 2: Generate Hypotheses

This hypothesis map enables faster prioritization during RCA by categorizing potential causes by origin

Internal Factors	External Factors
Algorithm update caused *low content diversity* or *poor relevance*	Major YouTube Shorts/Instagram Reels push
UX rollout (e.g., new autoplay, feed redesign) confused users	Viral challenge diverted attention elsewhere
A/B test configuration exposed wrong variant broadly	Regional bans or App Store issues
Infra or CDN latency degraded feed load times	PR incident or trust & safety failure (e.g., creator ban backlash)
Content moderation or removal led to stale feeds	

Example:

Internal Factors	External Factors
Algorithm update: low diversity, poor relevance	Competitor launch (Reels/Shorts)
A/B config errors (wrong variant exposed)	PR backlash or viral creator ban
UX rollout confusion (e.g., autoplay)	Regional bans or App Store issues
Infra/CDN latency	Seasonal or cultural event impact

Submission by Sia Puthusseri

Content moderation overreach	Negative press, trust & safety incidents

Step 3: Validate with Data

1. Segment Analysis

- Breakdown metrics by platform, region, cohort, content type
- Identify where the decline is *most severe vs. stable*

2. Model & Infra Logs

- Pull model latency logs, content diversity scores
- Audit recent model weight changes and rollout versions

3. Content Supply

- Drop in uploads? Removal spike? Dominance of a stale content type?

4. User Sentiment

- Scan app reviews, Reddit, TikTok comments
- Look for complaints about irrelevant feeds, creator bans, or slow load

5. A/B Test Isolation

- Freeze experiments, check control vs. treatment
- Look for global flag changes mistakenly pushed

Step 4: Refine Hypotheses Based on Patterns

- **If only new users tanked**
  → Revisit onboarding and cold-start logic; the model may not be surfacing engaging content early enough for first-time users.

- **If Gen Z in US/UK dropped**
  → Investigate recent shifts in trending content categories or possible relevance decay. Model might be overly optimized for historical engagement patterns, not emerging interests.

- **If likes & shares dropped but views didn't**
  → Likely a **content quality issue**, not delivery. Explore if high-view content is low-engagement (e.g., clickbaity or repeated).

- **If infra logs show latency spikes**
  → Check video loading times and model inference delays. Even a 200ms increase can affect FYP stickiness.

- **If watch time dipped but session starts remained stable**
  → Suggests declining **content retention**, not user acquisition. Validate diversity, freshness, and personalized flow.

- **If A/B test variants differ significantly in impact**
  → Isolate cohort exposure, check for **sample contamination**, feature flag misrouting, or

overlapping experiment noise.

- **If metrics dropped post model rollout but didn't recover on rollback**
  → Indicates possible **data leakage or feedback loops** already in effect. Further analysis needed on training data recency and cyclic dependencies.

- **If metrics dipped on one platform (e.g., Android only)**
  → Likely a **release-specific bug**, version compatibility issue, or device-specific regression. Cross-check app versions and crash logs.

- **If creators' engagement plummeted but viewers didn't**
  → Suspect shadow bans, moderation changes, or visibility throttling affecting content supply more than demand.

- **If drop coincides with geopolitical/regional events**
  → External factor likely. Validate by comparing with peer apps or regional news trends. Delay reactive changes unless confirmed causal.

Step 5: Identify the Root Cause

Example Diagnosis: A recommendation algorithm update over-indexed on content freshness, reducing diversity. This led to repetitive low-engagement videos, especially in Gen Z cohorts. Retention, re-engagement, and share metrics dropped.

Next Steps:
- Roll back model weights in a shadow test
- Monitor cohort-based recovery in real-time
- Re-tune model with new diversity thresholds

Stakeholder Communication: To ensure transparency and cross-functional alignment, I would immediately communicate the root cause, rollback strategy, and recovery plan to leads from Engineering, Product, Trust & Safety, and Data. Daily updates via Slack and a shared tracker would help maintain visibility, coordinate parallel fixes if needed, and reassure leadership.

Step 6: Act & Prevent

- **Immediate Fix:** Patch or rollback the change to stabilize user experience
- **Short-Term:** Tighten rollout gates, enable alerts on core KPIs (DAU, Watch Time)
- **Long-Term:**
  - Build ML-based anomaly detection for key metrics
  - Add feedback loops from creator dashboards & viewer sentiment
  - Implement pre-launch regression testing for ranking models

Real-World Take: If this was caused by a model regression, I'd immediately roll it back in a shadow test while monitoring engagement metrics like rewatch rate and content diversity scores segmented by user cohort. Recovery should show up within the next 6 hours if that was the root cause.

2a. You are in charge of serving high quality content to users on Tiktok Feed. What will you build?

1. Understanding the Problem

Users Impacted:

- New users (who represent ~30% of weekly app downloads) often churn within the first 7 days if content feels repetitive or low-value
- High-quality creators (especially new or niche) struggle to gain traction without early engagement, limiting their motivation and platform diversity
- Existing users experience fatigue from clickbait loops and recycled trends, reducing session depth and long-term satisfaction

What Does "High-Quality" Mean on TikTok? As a Product Manager, defining quality is essential for effective algorithmic design. High-quality content on TikTok exhibits the following attributes:

1. **Originality & Creativity:** Fresh, engaging ideas or unique spins on trends; avoids reposts and watermark reuse
2. **Strong Engagement & Retention:** High completion rates, replays, saves, comments, indicating meaningful interaction
3. **Content Safety & Compliance:** No guideline violations; family-safe, non-exploitative content
4. **Creator Consistency & Reputation:** A proven track record of valuable, enriching, and safe content from the creator
5. **Completeness & Value:** Videos that fulfill their promise (e.g., a full recipe, a clear tip) and don't rely on misleading tactics like baiting for part 2
6. **Relevance to Audience:** Content tailored to niche communities or specific interests rather than generic virality

These characteristics align with TikTok's mission to *inspire creativity and bring joy*, ensuring the For You feed highlights content that is enriching, not just attention-grabbing.

Specific Challenges:

- TikTok's recommender system prioritizes engagement (e.g., likes, watch time), which often benefits low-effort, trend-chasing content.
- High-quality, original content may go unnoticed without initial traction.
- Viewers experience fatigue due to repetitive, incomplete, or misleading content.
- Lack of dynamic response to users' real-time intent or mood

2. Proposed Solution: Quality-Optimized Recommender System

I propose a 3-layered recommender system to surface high-quality content by introducing a machine-learned Quality Score, session-level intent modeling, and dynamic boosting. This would improve ***user retention*** and ***uplift lesser-known creators*** while preserving personalization.

2a. You are in charge of serving high quality content to users on Tiktok Feed. What will you build?

1. Introduce Explicit Quality Ranking Signals

Beyond engagement metrics, each video would be scored on production clarity, originality, and completeness. A machine-learned **"Quality Score"** would leverage computer vision and NLP to detect high-resolution visuals, absence of watermarks, originality of ideas, and content that delivers a clear takeaway. This would allow well-crafted videos to be prioritized even before user engagement data is available.

2. Session-Level Intent Modeling

The feed would adapt in real time to reflect a user's current mood or context. For example:

- A user watching multiple recipe videos would trigger a temporary content shift toward cooking.
- Rapid scrolling signals disinterest, prompting feed diversification.
- The system models "intent vectors" for each session, temporarily re-weighting relevant content categories and quality filters.

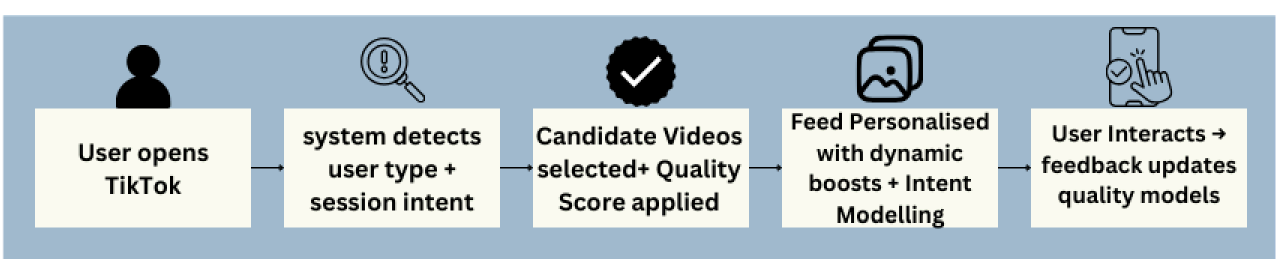
3. Dynamic Quality Boosting and Diversity Mix

A "Quality Boost Multiplier" would dynamically adjust ranking weight based on user type:

- **New users** receive feeds seeded with widely appealing high-quality content to build strong first impressions.
- **Established users** get periodic "high-quality wildcards" to break echo chambers and surface fresh content outside their niche.
- **Emerging creators** with well-produced videos and positive early feedback (e.g. saves, shares, follows) would receive accelerated distribution to avoid being overshadowed by popular creators.

Together, these strategies shift TikTok's "For You" feed toward consistently enriching, safe, and valuable content without compromising personalization or virality.

User Flow Example:



3. MVP: What We Launch First

To validate this solution with minimal engineering lift, our MVP includes:

- Quality Score prototype applied to a small batch of videos using existing metadata and NLP/CV models
- Session-level intent tagging from recent watch history (no new UI, just backend logic)
- Boosting logic for new users (applying high-quality wildcards to improve first-session experience)

This MVP can be A/B tested with low-risk cohorts and analyzed for improvement in quality-adjusted watch time and follow-through rates.

4. Implementation Feasibility: No New Systems, Just Smarter Layers

System Layer	What's going to Added	Why It's Feasible
Candidate Selection	Quality Score (CV + NLP model on metadata)	Leverages TikTok's existing ML model training and inference pipeline
Ranking Layer	Session-level intent vector & quality boost multiplier	Minor adjustments to scoring weights; no new model deployment required
Creator Feedback Loop	Creator reputation score + dashboard view	Uses existing moderation and analytics APIs
User Feedback	Lightweight signals (e.g., rewatch rate, saves, "not interested")	Already tracked; just reweighted as quality training data

TikTok's feed is beloved because it's fast, fun, and tailored. But for many users, that experience is starting to feel noisy, repetitive, and unoriginal. My solution doesn't slow the feed down, it sharpens it. By rewarding meaningful content, surfacing emerging voices, and adapting to real-time intent, keeping TikTok joyful *and* sustainable.

5. Success Metrics

North Star: Quality Adjusted Engagement Rate (QAER): % of high-quality videos that retain, satisfy, and convert (e.g. follow/save)

Quantitative	Qualitative	Validation
• Increase in video completion and save-to-view ratio • Boost in follows per video and shares from high-quality creators • Reduction in "Not Interested" flags and skip-through rate	• Micro-surveys: "Did this content inspire or entertain you?" • Improved user satisfaction with feed content (measured via NPS or in-app prompts)	• A/B test quality-enhanced algorithm vs. control group for uplift in watch time, repeat engagement, and satisfaction

2a. You are in charge of serving high quality content to users on Tiktok Feed. What will you build?

6. Trade-offs and Assumptions

Risks & Constraints:	Assumptions:	Edge Cases:
- Over-penalizing trendy content that lacks production polish - Bias in early machine predictions of quality - Added complexity in content scoring and moderation models	- Users want deeper, more satisfying experiences, not just fast entertainment - Creator reputation and production effort correlate with user value - Short-term loss in clickbait engagement is offset by long-term trust and retention	- Extremely niche content with low engagement but high satisfaction within small communities → **Solution:** Allow users to tag videos as "meaningful," "expert," or "underrated" to refine the feedback loop. Additionally, implement a lightweight "community upvote" feature during early test phases to identify high-value niche content outside traditional engagement models.

7. Rollout Plan & GTM Stratergy

Stage 1: A/B Testing

- Launch test to a subset of new users with high-quality feed prioritization
- Measure repeat session rate, average session duration, satisfaction prompts

→

Stage 2: Broader Experimentation

- Extend to mid-tier creators and content discovery pipelines
- Introduce creator dashboard for feedback on quality ranking and integrity score

→

Stage 3: Full Integration

- Bake quality scoring into baseline recommendation logic
- Continually improve ML models using human feedback, content audits, and creator input

Prioritization Strategy:

- Focus on onboarding experience and new creator uplift
- Use "standout content" thresholds to introduce high-quality wildcards into all user feeds

Target Segment:

- Mid-tier creators with strong creative output but limited visibility. This group will help validate the algorithm's ability to surface quality over virality.

Launch Approach:

- **Phase 1:** Invite-only "Creator Spotlight" campaign featuring top-performing creators based on Quality Score
- **Phase 2:** Feed banners and in-app prompts to educate users on the shift toward meaningful content
- **Phase 3:** Public rollout with creator-facing dashboards to visualize Quality Score and viewer feedback

Channels & Tactics:

- In-app banners on For You feed
- Targeted push notifications to engaged creators
- Integration with Creator Academy and community newsletters
- Social media campaigns highlighting creators who've benefited

Positioning & Messaging:

- Framed as a platform-wide shift to "Elevate What Matters" reinforcing TikTok's mission to surface content that inspires creativity, adds value, and sparks joy. The campaign celebrates creators who consistently deliver meaningful, complete, and safe content.

Submission by Sia Puthusseri

Appendix

Sources:

- TikTok Help Center – *How TikTok Recommends Content*newsroom.tiktok.comnewsroom.tiktok.comnewsroom.tiktok.com
- TikTok Newsroom – *How TikTok Recommends Videos #ForYou* (Jun 2020)newsroom.tiktok.com
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- TikTok Help Center – *How TikTok Shop Recommends Content* support.tiktok.comsupport.tiktok.com
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- Buffer Blog – *TikTok Algorithm Guide 2025*

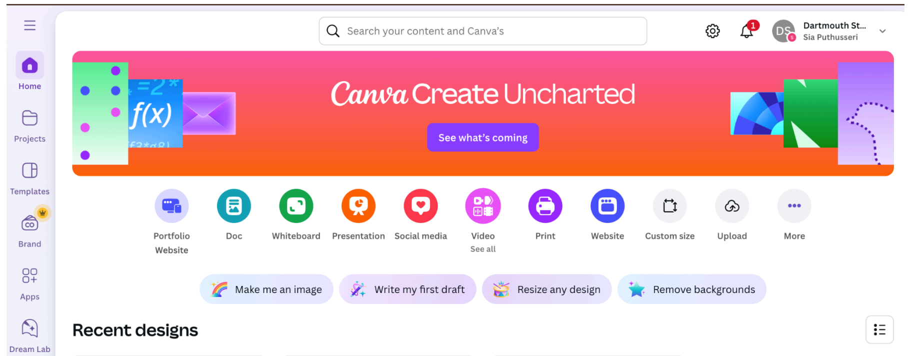
Sample User Journey: Maya, 17 – Student Viewer

Maya is a high school student who uses TikTok daily to relax between classes. She's curious, loves learning, and often looks for content that inspires or entertains but lately, her feed feels repetitive and shallow.

Wants / Desires	Inhibitors / Pain Points	How Quality Feed Helps
Discover videos that teach or inspire	Gets stuck in a loop of dance trends and clickbait edits	Quality Score prioritizes videos with originality, clarity, and value
Feel seen in the feed — not just part of a trend funnel	Feeds often misalign with her mood or curiosity	Session-Level Intent Modeling detects short-term interest spikes (e.g., DIY, mental health)
Find new creators who feel authentic and relatable	Same big creators dominate the feed	Dynamic Quality Boosting lifts emerging creators with strong early feedback
Avoid mindless scrolling and dopamine fatigue	Swipe loops leave her feeling drained	Quality-forward content design increases rewatch, save, and satisfaction moments
Before: Maya opens TikTok and scrolls through a few loud trends, swipes quickly, and closes the app after 4 minutes. **After:** Maya sees a mix of calm, high-quality content based on her recent behavior, saves a new favorite creator's video, and ends her session feeling satisfied — not overstimulated.		

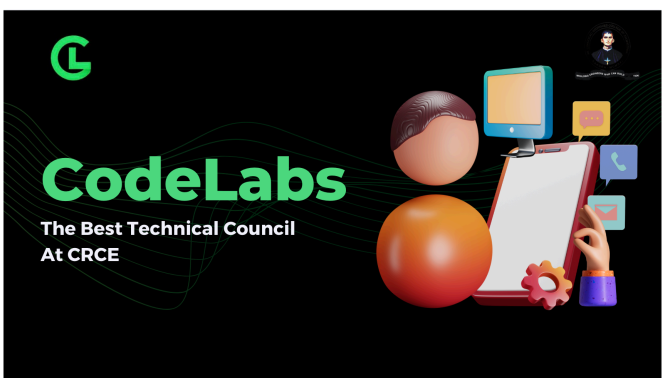
2b. What is your favorite product? Why? If you were to improve something on that product , what would it be and why?

Product Overview:
Canva is a cloud-based design platform that enables users to create high-quality visual content presentations, social media graphics, videos, and more without needing prior design expertise.



Why I Love Canva — A Personal Story

Canva has been part of my journey for as long as I can remember. It all started when I was helping my mom grow her small cake business. I wanted to support her by making her Instagram page look more professional, so I turned to Canva to create cute, engaging visuals and stories. At the time, I didn't have any design background, but Canva made it feel accessible. I could express creativity without being held back by complex tools or the need to code.



As I moved into undergrad, Canva became my default design partner whether it was building clean, compelling presentations for class, designing posters for my tech club, or creating interactive stories to boost social engagement. It gave me the tools to simplify complex ideas into clear, visually intuitive content. That's what truly drew me to Canva: *the ability to make your thinking visible in a way that resonates with others.*

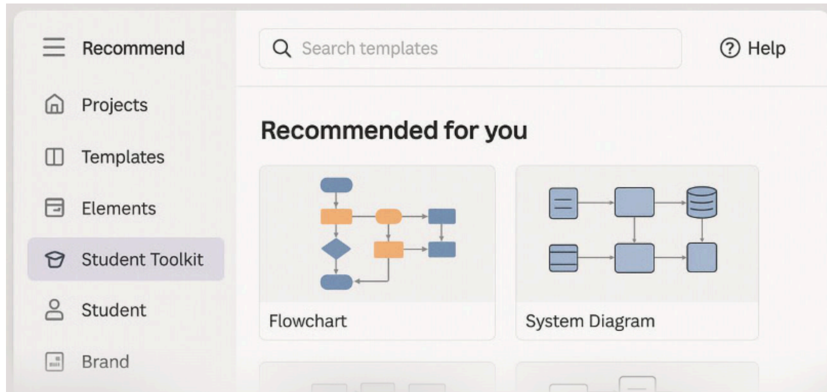
The platform's simplicity, combined with the power to create something that looks polished and thoughtful, made me realize how impactful great design can be. It's this experience that ultimately inspired me to pursue a UI/UX design internship. Canva helped me understand the value of good design not just as an aesthetic choice, but as a communication tool.

Opportunities for Improvement (based on personal experience):

2b. What is your favorite product? Why? If you were to improve something on that product , what would it be and why?

1. Expand academic and technical features offerings:

→***Challenge:*** As a student in a technical program, I frequently need to create flowcharts, ER diagrams, system architecture visuals, and data workflows. Canva doesn't currently offer built-in tools or templates for these, which often forces me to use external platforms like Lucidchart or draw.io. This breaks the creative flow and takes away from the "all-in-one" experience that Canva is otherwise so good at delivering.

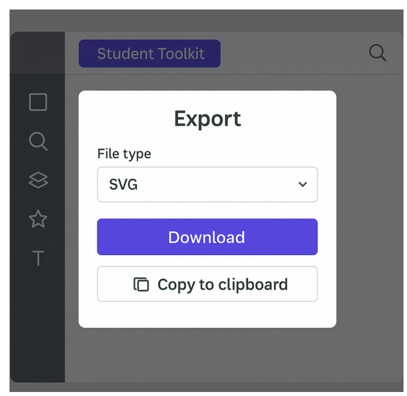


→ ***Recommendation:*** Introduce a **"Student Toolkit"** with technical diagrams, flowcharting tools, and templates tailored to academic presentations.

2. Enable flexible exporting and interoperability

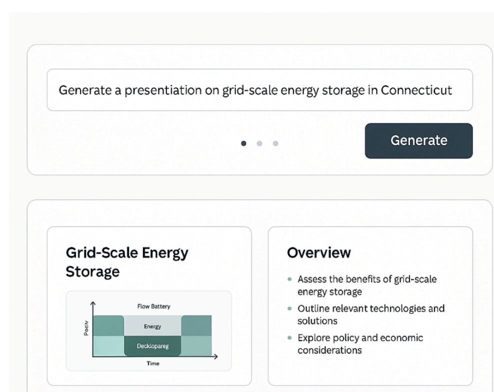
→***Challenge:*** As someone who creates custom graphics in Canva for web and app designs, I often hit a limitation, I can't copy elements into Figma or export them as individual SVGs. This makes it hard to fine-tune details like color or alignment, which are crucial during design handoffs.

→ ***Recommendation:*** Enable **advanced export options**, including SVG and component-level downloads, and add **clipboard-level interoperability** with design tools like Figma or Adobe XD. This would open the door for more intricate use cases and make Canva more appealing for designers who want to bridge quick mockups with detailed workflows.



3. Unlock full AI-powered presentation generation

→***Challenge:*** While Canva has made great progress with tools like Magic Write and AI-generated imagery, it still doesn't support end-to-end AI-powered presentation creation. Today, there are platforms that allow you to generate an entire presentation including structure, content, and design based on a single prompt. For a product that's known for intuitive, fast design, this feels like a missed opportunity.



→ ***Recommendation:*** Go beyond Magic Write by offering full AI-generated deck creation. A feature where users can input a prompt and get a fully structured, themed presentation would provide huge value, especially for time-constrained users like students and early-career professionals.

Submission by Sia Puthusseri